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United States Patent	12419149
Kind Code	B2
Date of Patent	September 16, 2025
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Circuit board having multiple solder resists and displaying apparatus having the same

Abstract

A circuit board includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections and defining a pad open region exposing portions of the interconnections, a second PSR covering the first PSR and having an opening exposing the pad open region. The opening of the second PSR is larger than the pad open region of the first PSR.

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Appl. No.: **17/684970**

Filed: **March 02, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220285599 A1	Sep. 08, 2022

Related U.S. Application Data

us-provisional-application US 63310367 20220215
us-provisional-application US 63157180 20210305

Publication Classification

Int. Cl.: **H10H20/857** (20250101); **H01L25/075** (20060101)

U.S. Cl.:

CPC **H10H20/857** (20250101); **H01L25/0753** (20130101);

Field of Classification Search

CPC: H01L (33/62); H01L (25/075); H01L (25/0753); H01L (25/0756); H01L (25/167); H10K (59/60)

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS AND PRIORITY (1) The present application is a nonprovisional application which claims priority to and benefit of U.S. Provisional Applications Nos. 63/157,180 filed Mar. 5, 2021 and 63/310,367 filed Feb. 15, 2022, the disclosure of which are incorporated by reference in their entirety as if they are fully set forth herein.

TECHNICAL FIELD

(1) Exemplary embodiments relate to a micro LED displaying apparatus, and a circuit board for mounting micro LEDs and a displaying apparatus having the same.

BACKGROUND

(2) Light emitting devices are semiconductor devices using light emitting diodes which are inorganic light sources, and are used in various technical fields such as displaying apparatuses, automobile lamps, general lighting, and the like. Light emitting diodes have advantages such as longer lifespan, lower power consumption, and quicker response, than conventional light sources, and thus, the light emitting diodes have been replacing the conventional light sources.

(3) The conventional light emitting diodes have been generally used as backlight light sources in displaying apparatuses. However, displaying apparatuses that directly realize images using the light emitting diodes have been recently developed. Such displays are also referred to as micro LED displays.

(4) Micro LED displays can realize images of various colors by driving micro LEDs mounted on a circuit board. In general, the micro LED display includes a plurality of pixels in order to realize various images, each including sub-pixels corresponding to one of blue, green, and red light. As such, a color of a certain pixel is typically determined based on the colors of the sub-pixels, so that images can be realized through the combination of such pixels.

(5) Meanwhile, a dark color such as black is realized by turning off micro LEDs, and may be affected by a color of an upper surface of the circuit board observed from a screen side. In particular, when interconnections formed on the circuit board are observed, it is difficult to display desirable black color.

(6) The interconnections formed on the circuit board are generally covered with a solder resist such as a photosensitive solder resist (PSR), and, as a thickness of the PSR becomes thinner, the interconnections are easily observed. Meanwhile, it is possible to prevent the interconnections from being observed by forming a thick PSR covering the interconnections. However, when the PSR is formed thick, patterning for exposing pads of the circuit board is difficult, and an open region is not clearly defined. A pad open region on the circuit board is used as an arrangement mark to arrange the micro LEDs. However, if the pad open region on the circuit board is not clearly defined, it is difficult to accurately arrange the micro LEDs to the pad open region, and thus, a bonding failure is likely to occur.

SUMMARY

(7) Exemplary embodiments provide a circuit board capable of clearly defining a pad open region and preventing interconnections from being observed, and a displaying apparatus having the same.

(8) A circuit board according to one or more exemplary embodiments of the present disclosure includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections, and defining a pad open region exposing portions of the interconnections, a second PSR covering the first PSR, and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR.

(9) A displaying apparatus according to one or more exemplary embodiments of the present disclosure includes, a circuit board, a unit pixel disposed on the circuit board, and a molding member covering the unit pixels, in which the circuit board includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections, and defining a pad open region exposing portions of the interconnections, and a second PSR covering the first PSR and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR, and the unit pixel is disposed in the pad open region.
(10) A pixel module according to one or more exemplary embodiments of the present disclosure includes a circuit board, a unit pixel disposed on the circuit board, and a molding member covering the unit pixels, in which the circuit board includes: a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections and defining a pad open region exposing portions of the interconnections, and a second PSR covering the first PSR, and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR, and the unit pixel is disposed in the pad open region.
(11) According to one or more embodiments of the present disclosure, a circuit board includes a base, a plurality of interconnections disposed on an upper surface of the base, each interconnection comprising a plurality of portions, a first photosensitive solder resist (PSR) and a second PSR. The first PSR covers a portion of each of the plurality of interconnections, and the first PSR is arranged to define a pad open region configured to expose a remainder portion of each of the plurality of interconnection. The second PSR covers the first PSR and has an opening configured to expose the pad open region. The opening of the second PSR is larger than the pad open region of the first PSR.
(12) In at least one variant, the second PSR is thicker than the first PSR.
(13) In another variant, the first PSR has a thickness of about 10 μm to about 15 μm, and the second PSR has a thickness of about 20 μm to about 25 μm.
(14) In another variant, the plurality of interconnections includes four interconnections disposed on the upper surface of the base and the first PSR covers the portion of each of the four interconnections. The pad open region is further configured to expose the remainder portion of each of the four interconnections.
(15) In another variant, each of the plurality of interconnections includes a first portion located in the pad open region. A second portion is located in the opening of the second PSR and covered with the first PSR, and a third portion is covered with both the first PSR and second PSR. The remainder portion of each of the plurality of interconnections corresponds to the first portion and the portion of each of the plurality of interconnections, covered by the first PSR, correspond to the second portion and the third portion.
(16) In another variant, the circuit board is a printed circuit board.
(17) According to one or more embodiments of the present disclosure, a displaying apparatus includes a circuit board, a unit pixel disposed on the circuit board, and a molding member covering the unit pixel. The circuit board includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) and a second PSR. The first PSR covers a portion of each of the plurality of interconnection. The first PSR is arranged to define a pad open region configured to expose a remainder portion of each of the plurality of interconnections. The second PSR covers the first PSR and has an opening configured to expose the pad open region. The opening of the second PSR is larger than the pad open region of the first PSR. The unit pixel is disposed in the pad open region and in the opening of the second PSR such that the unit pixel is electrically coupled to the remainder portion of each of the plurality of interconnections in the pad open region.
(18) In at least one variant, the unit pixel includes at least three light emitting devices disposed adjacent to one another.

(19) In another variant, the unit pixel includes a plurality of light emitting stacks stacked one above another. A plurality of connection electrodes is electrically connected to the plurality of light emitting stacks.
(20) In another variant, the unit pixel is arranged to cover the pad open region.
(21) In another variant, the molding member includes a light absorbing material.
(22) In another variant, the displaying apparatus further includes a display panel and a pixel module disposed on the display panel. The pixel module includes the circuit board and a plurality of unit pixels.
(23) According to one or more embodiments of the present disclosure, a displaying apparatus includes a panel substrate and a plurality of pixel modules arranged on the panel substrate. Each pixel module includes a circuit board, a plurality of unit pixels disposed on the circuit board and including a selected unit pixel and a molding member covering the plurality of unit pixels. The circuit board further includes a base having a plurality of interconnections on an upper surface thereof. An interconnection of the plurality of interconnections includes a first part, a second part, and a third part. The circuit board further includes a pad open region formed in a first photosensitive solder resist (PSR) layer and an opening. The pad open region is configured to expose the first part of the interconnection. The first PSR layer covers the second part and the third part of the interconnection. The opening is formed in a second PSR layer covering a portion of the first PSR layer, and the pad open region is exposed in the opening of the second PSR layer. The selected unit pixel is disposed in the opening such that the selected unit pixel is electrically connected to the first part of the interconnection in the pad open region.
(24) In at least one variant, the selected unit pixel covers the pad open region upon being mounted in the opening.
(25) In another variant, the first part of the interconnection is located in the pad open region, the second part located in the opening of the second PSR layer and covered with the first PSR layer, and the third part covered with both the first PSR layer and the second PSR layer.
(26) In another variant, the selected unit pixel further includes a plurality of light emitting stacks stacked one above another on a substrate, and a plurality of connection electrodes electrically connected to the plurality of light emitting stacks.
(27) In another variant, the plurality of light emitting stacks is operable to emit light of different peak wavelengths from one another. One of the plurality of light emitting stacks, farther from the substrate, emits light of a longer wavelength than another light emitting stack near the substrate.
(28) In another variant, an emission area of each of the plurality of light emitting stacks increases as a distance to the substrate decreases.
(29) In another variant, the selected unit pixel further includes a plurality of connection electrodes which is electrically connected to the first part of the interconnection via a bonding material. The bonding material is disposed in the pad open region.
(30) In another variant, a portion of the first PSR layer exposed in the opening vertically overlaps with the selected unit pixel. At least portions of side surfaces of the selected unit pixel are located inside the opening of the second PSR layer such that when the selected unit pixel moves in a lateral direction while the selected unit pixel is mounted, a side surface of the selected unit pixel contacts a side wall of the opening of the second PSR layer, thereby preventing the selected unit pixel from being deviated from the opening.

Description

DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic plan view illustrating a displaying apparatus according to an exemplary embodiment.

(2) FIG. **2** is a schematic plan view illustrating a pixel module according to an exemplary embodiment.
(3) FIG. **3**A is a schematic plan view illustrating a light emitting device according to an exemplary embodiment.
(4) FIG. **3**B is a schematic cross-sectional view taken along line A-A′ of FIG. **3**A.
(5) FIG. **4**A is a schematic plan view illustrating a unit pixel according to an exemplary embodiment.
(6) FIG. **4**B is a schematic cross-sectional view taken along line B-B′ of FIG. **4**A.
(7) FIG. **4**C is a schematic cross-sectional view taken along line C-C′ of FIG. **4**A.
(8) FIG. **5**A is a partially enlarged plan view illustrating a portion of FIG. **2** so as to describe a pixel module according to an exemplary embodiment.
(9) FIG. **5**B is a schematic partially enlarged plan view illustrating a circuit board according to an exemplary embodiment.
(10) FIG. **5**C is a schematic cross-sectional view taken along line D-D′ of FIG. **5**A.
(11) FIG. **6**A is a schematic cross-sectional view illustrating a unit pixel according to another exemplary embodiment.
(12) FIG. **6**B is a schematic plan view illustrating a unit pixel according to another exemplary embodiment.
(13) FIG. **7**A is an image illustrating a pixel module according to the related art.
(14) FIG. **7**B is an image illustrating a pixel module according to an exemplary embodiment.
(15) FIG. **8**A is a schematic plan view illustrating a pixel module according to an exemplary embodiment.
(16) FIG. **8**B is a schematic rear view illustrating the pixel module according to FIG. **8**A.
(17) FIG. **8**C is a schematic circuit diagram illustrating the pixel module according to FIG. **8**A.
(18) FIG. **9**A is a schematic plan view illustrating a pixel module according to another exemplary embodiment.
(19) FIG. **9**B is a schematic rear view illustrating the pixel module according to FIG. **9**A.
(20) FIG. **9**C is a schematic circuit diagram illustrating the pixel module according to FIG. **9**A.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(21) Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The following embodiments are provided by way of example so as to fully convey the spirit of the present disclosure to those skilled in the art to which the present disclosure pertains. Accordingly, the present disclosure is not limited to the embodiments disclosed herein and can also be implemented in different forms. In the drawings, widths, lengths, thicknesses, and the like of elements can be exaggerated for clarity and descriptive purposes. When an element or layer is referred to as being “disposed above” or “disposed on” another element or layer, it can be directly “disposed above” or “disposed on” the other element or layer or intervening elements or layers can be present. Throughout the specification, like reference numerals denote like elements having the same or similar functions.
(22) A circuit board according to one or more exemplary embodiments of the present disclosure includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections and defining a pad open region exposing portions of the interconnections, and a second PSR covering the first PSR and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR.
(23) The second PSR may be thicker than the first PSR. In an exemplary embodiment, the first PSR may have a thickness of about 10 μm to about 15 μm, and the second PSR may have a thickness of about 20 μm to about 25 μm.
(24) The pad open region may expose respective portions of four interconnections.
(25) Each of the interconnections may include a first portion located in the pad open region, a

second portion located in the opening of the second PSR and covered with the first PSR, and a third portion covered with the first PSR and the second PSR.
(26) The circuit board may be a printed circuit board.
(27) A displaying apparatus according to one or more exemplary embodiments of the present disclosure includes a circuit board, a unit pixel disposed on the circuit board, and a molding member covering the unit pixels. The circuit board includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections and defining a pad open region exposing portions of the interconnections, and a second PSR covering the first PSR and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR, and the unit pixel is disposed in the pad open region.
(28) The second PSR may be thicker than the first PSR. In an exemplary embodiment, the first PSR may have a thickness of about 10 μm to about 15 μm, and the second PSR may have a thickness of about 20 μm to about 25 μm.
(29) The pad open region may expose respective portions of four interconnections.
(30) Each of the interconnections may include a first portion located in the pad open region, a second portion located in the opening of the second PSR and covered with the first PSR, and a third portion covered with the first PSR and the second PSR.
(31) The circuit board may be a printed circuit board.
(32) In some exemplary embodiments, the unit pixel may include at least three light emitting devices disposed adjacent to one another.
(33) In other exemplary embodiments, the unit pixel may include a plurality of light emitting stacks stacked one above another; and connection electrodes electrically connected to the light emitting stacks.
(34) The unit pixel may cover the pad open region.
(35) The molding member may include a light absorbing material.
(36) The displaying apparatus may further include a display panel and a pixel module disposed on the display panel, and the pixel module may include the circuit board and the unit pixel.
(37) A pixel module according to one or more exemplary embodiments of the present disclosure includes a circuit board, a unit pixel disposed on the circuit board, and a molding member covering the unit pixels. The circuit board includes a base having a plurality of interconnections on an upper surface thereof, a first photosensitive solder resist (PSR) covering the interconnections and defining a pad open region exposing portions of the interconnections, and a second PSR covering the first PSR and having an opening exposing the pad open region, in which the opening of the second PSR is larger than the pad open region of the first PSR, and the unit pixel is disposed in the pad open region.
(38) The unit pixel may cover the pad open region.
(39) Each of the interconnections may include a first portion located in the pad open region, a second portion located in the opening of the second PSR and covered with the first PSR, and a third portion covered with the first PSR and the second PSR.
(40) Hereinafter, exemplary embodiments will be described in detail with reference to the accompanying drawings.
(41) FIG. **1** is a schematic plan view illustrating a displaying apparatus **10000** according to an exemplary embodiment, and FIG. **2** is a schematic plan view illustrating a pixel module **1000** according to an exemplary embodiment.
(42) Referring to FIGS. **1** and **2**, the displaying apparatus **10000** may include a panel substrate **2100** and a plurality of pixel modules **1000**.
(43) The displaying apparatus **10000** is not particularly limited, but it may include a virtual reality (VR) displaying apparatus such as a micro LED TV, a smart watch, a VR headset, or an augment reality (AR) displaying apparatus such as augmented reality glasses.

(44) The panel substrate **2100** may include a circuit for a passive matrix driving or active matrix driving manner. In an exemplary embodiment, the panel substrate **2100** may include interconnections and resistors therein, and, in another exemplary embodiment, the panel substrate **2100** may include interconnections, transistors, and capacitors. The panel substrate **2100** may also have pads that are capable of being electrically connected to the disposed circuit on an upper surface thereof.
(45) In an exemplary embodiment, the plurality of pixel modules **1000** is arranged on the panel substrate **2100**. Each of the pixel modules **1000** may include a circuit board **1001**, a plurality of unit pixels **100** disposed on the circuit board **1001**, and a molding member **200** covering the unit pixels **100**. In another exemplary embodiment, the plurality of unit pixels **100** may be arranged directly on the panel substrate **2100**, and the molding member **200** may cover the unit pixels **100**.
(46) Each of the unit pixels **100** includes a plurality of light emitting devices **10***a*, **10***b*, and **10***c*. The light emitting devices **10***a*, **10***b*, and **10***c* may emit light of different colors from one another. The light emitting devices **10***a*, **10***b*, and **10***c* in each of the unit pixels **100** may be arranged in a line, as illustrated in FIG. **2**. In an exemplary embodiment, the light emitting devices **10***a*, **10***b*, and **10***c* may be arranged in a vertical direction with respect to a display screen on which an image is implemented. The inventive concepts are not limited thereto, and for instance, the light emitting devices **10***a*, **10***b*, and **10***c* may be arranged in a lateral direction or diagonally with respect to the display screen on which the image is implemented.
(47) While the light emitting devices **10***a*, **10***b*, and **10***c* are mounting directly on the panel substrate **2100**, a mounting failure of a few light mounting devices may occur. All of the light emitting devices and the panel substrate **2100** may be affected by the mounting failure, leading to increased cost. Alternatively, it is possible to select desirable unit pixels **100**, after manufacturing unit pixels **100** on which the light emitting devices **10***a*, **10***b*, and **10***c* are mounted, and mount selected unit pixels **100** on the panel substrate **2100**. Cost loss may be reduced or avoided.
(48) Hereinafter, each element of the displaying apparatus **10000** will be described in detail in an order of the light emitting devices **10***a*, **10***b*, and **10***c*, the unit pixel **100**, and the pixel module **1000** disposed in the displaying apparatus **10000**.
(49) First, FIG. **3**A is a schematic plan view illustrating the light emitting device **10***a* according to an exemplary embodiment, and FIG. **3**B is a schematic cross-sectional view taken along line A-A′ of FIG. **3**A. Herein, the light emitting device **10***a* is exemplarily described, but as the light emitting devices **10***b* and **10***c* have a substantially similar structure to that of the light emitting device **10***a*, repeated descriptions thereof will be omitted.
(50) Referring to FIGS. **3**A and **3**B, the light emitting device **10***a* may include a light emitting structure including a first conductivity type semiconductor layer **21**, an active layer **23**, and a second conductivity type semiconductor layer **25**, an ohmic contact layer **27**, a first contact pad **53**, a second contact pad **55**, an insulation layer **59**, a first electrode pad **61**, and a second electrode pad **63**.
(51) The light emitting device **10***a* may have a rectangular shape having a major axis and a minor axis in plan view. For example, a length of the major axis may have a size of about 100 μm or less, and a length of the minor axis may have a size of about 70 μm or less. In some forms, the light emitting devices **10***a*, **10***b*, and **10***c* may have substantially similar shapes and sizes, but the present disclosure is not limited thereto.
(52) The light emitting structure, that is, the first conductivity type semiconductor layer **21**, the active layer **23**, and the second conductivity type semiconductor layer **25** may be grown on a substrate. The substrate may be one of various substrates that are used to grow semiconductors, such as a gallium nitride substrate, a GaAs substrate, a Si substrate, a sapphire substrate, especially a patterned sapphire substrate. The growth substrate may be separated from the semiconductor layers using a process such as a mechanical grinding, a laser lift off, a chemical lift off process, or the like. However, the inventive concepts are not limited thereto, and, in some exemplary

embodiments, a portion of the substrate may remain to form at least a portion of the first conductivity type semiconductor layer **21**.

(53) In an exemplary embodiment, in a case of the light emitting device **10***a* emitting red light, the semiconductor layers may include aluminum gallium arsenide (AlGaAs), gallium arsenide phosphide (GaAsP), aluminum gallium indium phosphide (AlGaInP), or gallium phosphide (GaP).

(54) In a case of the light emitting device **10***b* emitting green light, the semiconductor layers may include indium gallium nitride (InGaN), gallium nitride (GaN), gallium phosphide (GaP), aluminum gallium indium phosphide (AlGaInP), or aluminum gallium phosphide (AlGaP).

(55) In an exemplary embodiment, in a case of the light emitting device **10***c* emitting blue light, the semiconductor layers may include gallium nitride (GaN), indium gallium nitride (InGaN), or zinc selenide (ZnSe).

(56) The first conductivity type and the second conductivity type have opposite polarities, such as when the first conductivity type is an n-type, the second conductivity type becomes a p-type, or when the first conductivity type is a p-type, the second conductivity type becomes an n-type.

(57) The first conductivity type semiconductor layer **21**, the active layer **23**, and the second conductivity type semiconductor layer **25** may be grown on the substrate in a chamber using a known process such as metal organic chemical vapor deposition (MOCVD) process. In addition, the first conductivity type semiconductor layer **21** includes n-type impurities (e.g., Si, Ge, and Sn), and the second conductivity type semiconductor layer **25** includes p-type impurities (e.g., Mg, Sr, and Ba). In a case of the light emitting device **10***b* or **10***c* emitting green light or blue light, the first conductivity type semiconductor layer **21** may include GaN or AlGaN containing Si as a dopant, and the second conductivity type semiconductor layer **25** may include GaN or AlGaN containing Mg as a dopant.

(58) Although the first conductivity type semiconductor layer **21** and the second conductivity type semiconductor layer **25** are shown as single layers in FIGS. **3**A-**3**B, these layers may be multiple layers, and may also include a superlattice layer. The active layer **23** may include a single quantum well structure or a multiple quantum well structure, and a composition ratio of a compound semiconductor may be adjusted to emit light with a desired wavelength. For example, the active layer **23** may emit blue light, green light, red light, or ultraviolet light.

(59) The second conductivity type semiconductor layer **25** and the active layer **23** may have a mesa M structure and may be disposed on the first conductivity type semiconductor layer **21**. The mesa M may include the second conductivity type semiconductor layer **25** and the active layer **23**, and may include a portion of the first conductivity type semiconductor layer **21** as shown in FIG. **3**B. The mesa M is located on a partial region of the first conductivity type semiconductor layer **21**, and an upper surface of the first conductivity type semiconductor layer **21** may be exposed around the mesa M.

(60) In the illustrated exemplary embodiment, the mesa M is formed so as to expose the first conductivity type semiconductor layer **21** around the mesa M. In another exemplary embodiment, a through hole may be formed through the mesa M to expose the first conductivity type semiconductor layer **21**.

(61) In an exemplary embodiment, the first conductivity type semiconductor layer **21** may have a flat light exiting surface. In another exemplary embodiment, the first conductivity type semiconductor layer **21** may have a concave-convex pattern formed by surface texturing on a side of the light exiting surface. Surface texturing may be carried out by patterning, for example, using a dry or wet etching process. For example, cone-shaped protrusions (not shown) may be formed on the light exiting surface of the first conductivity type semiconductor layer **21**, a height of the cone may be about 2 μm to about 3 μm, a distance between the cones may be about 1.5 μm to about 2 μm, and a diameter of a bottom of the cone may be about 3 μm to about 5 μm. The cone may also be truncated, in which an upper diameter of the cone may be about 2 μm to about 3 μm.

(62) In another exemplary embodiment, the concave-convex pattern may include a first concave-

convex pattern and a second concave-convex pattern additionally formed on the first concave-convex pattern.
(63) By forming the concave-convex pattern on the surface of the first conductivity type semiconductor layer **21**, total internal reflection may be reduced, thereby increasing light extraction efficiency. Surface texturing may be carried out on the first conductivity type semiconductor layers of all of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*, and viewing angles of light emitted from the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may become uniform. However, the inventive concepts are not limited thereto, and at least one of the light emitting devices **10***a*, **10***b*, and **10***c* may have a flat surface without a concave-convex pattern.
(64) The ohmic contact layer **27** is disposed on the second conductivity type semiconductor layer **25** to be in ohmic contact with the second conductivity type semiconductor layer **25**. The ohmic contact layer **27** may be formed of a single layer or multiple layers, and may be formed of a transparent conductive oxide film or a metal film. For example, the transparent conductive oxide film may include ITO, ZnO, or the like, and the metal film may include a metal such as Al, Ti, Cr, Ni, Au, or the like and alloys thereof.
(65) The first contact pad **53** is disposed on the exposed first conductivity type semiconductor layer **21**. The first contact pad **53** may be in ohmic contact with the first conductivity type semiconductor layer **21**. For example, the first contact pad **53** may be formed of an ohmic metal layer (not shown) in ohmic contact with the first conductivity type semiconductor layer **21**. The ohmic metal layer of the first contact pad **53** may be appropriately selected depending on a semiconductor material of the first conductivity type semiconductor layer **21**. The first contact pad **53** may be omitted.
(66) The second contact pad **55** may be disposed on the ohmic contact layer **27**. The second contact pad **55** is electrically connected to the ohmic contact layer **27**. The second contact pad **55** may be omitted.
(67) The insulation layer **59** covers the mesa M, the ohmic contact layer **27**, the first contact pad **53**, and the second contact pad **55**. The insulation layer **59** has openings **59***a* and **59***b* exposing the first contact pad **53** and the second contact pad **55**. The insulation layer **59** may be formed as a single layer or multiple layers. Further, the insulation layer **59** may include a distributed Bragg reflector in which insulation layers having different refractive indices from one another are stacked. For example, the distributed Bragg reflector may include at least two insulation layers selected from SiO_2, Si_3N_4, SiON, TiO_2, Ta_2O_5, and Nb_2O_5.
(68) The distributed Bragg reflector reflects light emitted from the active layer **23**. The distributed Bragg reflector may exhibit high reflectance over a relatively wide wavelength range including a peak wavelength of light emitted from the active layer **23**, and may be designed in consideration of an incident angle of light. In an exemplary embodiment, the distributed Bragg reflector may have a higher reflectance for light incident at an incident angle of 0 degrees than that for light incident at a different incident angle. In another exemplary embodiment, the distributed Bragg reflector may have a higher reflectance for light incident at a particular incident angle than that for light incident at the incident angle of 0 degrees. For example, the distributed Bragg reflector may have a higher reflectance for light incident at an incident angle of 10 degrees than that for light incident at the incident angle of 0 degrees.
(69) Meanwhile, the light emitting structure of the blue light emitting device **10***c* has higher internal quantum efficiency compared to those of the light emitting structures of the red light emitting device **10***a* and the green light emitting device **10***b*. Accordingly, the blue light emitting device **10***c* may exhibit higher light extraction efficiency than those of the red and green light emitting devices **10***a* and **10***b*. As such, it may be difficult to properly maintain a color mixing ratio of red light, green light, and blue light.
(70) To adjust the color mixing ratio of red light, green light, and blue light, the distributed Bragg reflectors applied to the light emitting devices **10***a*, **10***b*, and **10***c* may be formed to have different reflectance from one another. For example, the blue light emitting device **10***c* may have the

distributed Bragg reflector having a relatively low reflectance compared to those of the red and green light emitting devices **10***a* and **10***b*. For example, the distributed Bragg reflector formed in the blue light emitting device **10***c* may have a reflectance of 95% or less at the incident angle of 0 degrees for blue light generated in the active layer **23**, and further 90% or less, the distributed Bragg reflector formed in the green light emitting device **10***b* may have a reflectance of about 95% or more and 99% or less at the incident angle of 0 degrees for green light, and the distributed Bragg reflector formed in the red light emitting device **10***a* may have a reflectance of 99% or more at the incident angle of 0 degrees for red light.

(71) In an exemplary embodiment, the distributed Bragg reflectors applied to the red, green, and blue light emitting devices **10***a*, **10***b*, and **10***c* may have a substantially similar thickness. For example, a difference in thickness between the distributed Bragg reflectors applied to these light emitting devices **10***a*, **10***b*, and **10***c* may be 10% or less of a thickness of a thickest distributed Bragg reflector. By reducing the thickness difference between the distributed Bragg reflectors, process conditions applied to the red, green, and blue light emitting devices **10***a*, **10***b*, and **10***c*, for example, a process of patterning the insulation layer **59**, may be similarly set, and furthermore, the unit pixel manufacturing process can be simplified. Moreover, the distributed Bragg reflectors applied to the red, green, and blue light emitting devices **10***a*, **10***b*, and **10***c* may have a substantially similar stacking number. However, the inventive concepts are not limited thereto.

(72) The first electrode pad **61** and the second electrode pad **63** are disposed on the insulation layer **59**. The first electrode pad **61** may extend from an upper region of the first contact pad **53** to an upper region of the mesa M, and the second electrode pad **63** may be disposed in the upper region of the mesa M. The first electrode pad **61** may be connected to the first contact pad **53** through the opening **59***a*, and the second electrode pad **63** may be electrically connected to the second contact pad **55**. The first electrode pad **61** may be directly in ohmic contact with the first conductivity type semiconductor layer **21**, and in this case, the first contact pad **53** may be omitted. In addition, when the second contact pad **55** is omitted, the second electrode pad **63** may be directly connected to the ohmic contact layer **27**.

(73) The first and/or second electrode pads **61** and **63** may be formed of a single layer or a multilayer metal. As a material of the first and/or second electrode pads **61** and **63**, metals such as Al, Ti, Cr, Ni, Au, or the like and alloys thereof may be used. For example, the first and second electrode pads **61** and **63** may include a Ti layer or a Cr layer as an upper most layer, and an Au layer thereunder.

(74) Although the light emitting device **10***a* according to the exemplary embodiment has been briefly described with reference to FIGS. **3**A-**3**B, the light emitting device **10***a* may further include a layer having additional functions in addition to the above-described layers. For example, various layers such as a reflection layer for reflecting light, an additional insulation layer for insulating a specific element, and a solder preventing layer for preventing diffusion of solder may be further included.

(75) In addition, when a flip chip type light emitting device is formed, the mesa may be formed to have various shapes, and locations and shapes of the first and second electrode pads **61** and **63** may also be variously modified. Further, the ohmic contact layer **27** may be omitted, and the second contact pad **55** or the second electrode pad **63** may directly contact the second conductivity type semiconductor layer **25**.

(76) FIG. **4**A is a schematic plan view illustrating a unit pixel **100** according to an exemplary embodiment, FIG. **4**B is a schematic cross-sectional view taken along line B-B′ of FIG. **4**A, and FIG. **4**C is a schematic cross-sectional view taken along line C-C′ of FIG. **4**A.

(77) Referring to FIGS. **4**A, **4**B, and **4**C, the unit pixel **100** may include a transparent substrate **121**, first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*, a surface layer **122**, a light blocking layer **123**, an adhesive layer **125**, a step adjustment layer **127**, connection layers **129***a*, **129***b*, **129***c*, and **129***d*, and an insulation material layer **131**.

(78) The unit pixel **100** provides a single pixel including the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*. The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* emit light of different colors, and the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* correspond to subpixels, respectively.
(79) The transparent substrate **121** is a light transmissive substrate such as PET, glass substrate, quartz, sapphire substrate, or the like. The transparent substrate **121** is disposed on a light exiting surface of the displaying apparatus (**10000** in FIG. **1**), and light emitted from the light emitting devices **10***a*, **10***b*, and **10***c* is emitted to the outside through the transparent substrate **121**. The transparent substrate **121** may have an upper surface and a lower surface. The transparent substrate **121** may include a concave-convex pattern **121***p* on a surface facing the light emitting devices **10***a*, **10***b*, and **10***c*, that is, the upper surface. The concave-convex pattern **121***p* scatters light emitted from the light emitting devices **10***a*, **10***b*, and **10***c* to increase a viewing angle. In addition, light emitted from the light emitting devices **10***a*, **10***b*, and **10***c* having different viewing angle characteristics from one another may be emitted at a uniform viewing angle by the concave-convex pattern **121***p*. As such, it is possible to prevent an occurrence of color difference depending on the viewing angle.
(80) The concavo-convex pattern **121***p* may be regular or irregular. The concavo-convex pattern **121***p* may have a pitch of about 3 μm, a diameter of about 2.8 μm, and a height of about 1.8 μm, for example. The concavo-convex pattern **121***p* may be a pattern generally applied to a patterned sapphire substrate, but the inventive concepts are not limited thereto.
(81) The transparent substrate **121** may also include an anti-reflection coating, may include an anti-glare layer, or may be treated with an anti-glare treatment. The transparent substrate **121** may have a thickness of about 50 μm to about 300 μm for example.
(82) The transparent substrate **121** is disposed on the light exiting surface and does not include a circuit. However, the inventive concepts are not limited thereto, and, in some exemplary embodiments, the transparent substrate **121** may include the circuit.
(83) Although a single unit pixel **100** is illustrated to be formed on a single transparent substrate **121**, a plurality of unit pixels **100** may be formed on the single transparent substrate **121**.
(84) The surface layer **122** covers the concave-convex pattern **121***p* of the transparent substrate **121**. The surface layer **122** may be formed along a shape of the concave-convex pattern **121***p*. The surface layer **122** may improve adhesion of the light blocking layer **123** formed thereon. For example, the surface layer **122** may be formed of a silicon oxide layer. The surface layer **122** may be omitted depending on a type of the transparent substrate **121**.
(85) The light blocking layer **123** is formed on the upper surface of the transparent substrate **121**. The light blocking layer **123** may contact the surface layer **122**. The light blocking layer **123** may include an absorbing material which absorbs light such as carbon black. The light absorbing material may prevent light generated in the light emitting devices **10***a*, **10***b*, and **10***c* from leaking from a region between the transparent substrate **121** and the light emitting devices **10***a*, **10***b*, and **10***c* toward a side surface thereof, and may improve contrast of the displaying apparatus.
(86) The light blocking layer **123** may have windows **123***a*, **123***b*, and **123***c* for a light path, so that light generated in the light emitting devices **10***a*, **10***b*, and **10***c* is incident on the transparent substrate **121**, and for this purpose, the light blocking layer **123** on the transparent substrate **121** may be patterned so as to expose the transparent substrate **121**. Widths of the windows **123***a*, **123***b*, and **123***c* may be narrower than those of the light emitting devices, but the inventive concepts are not limited thereto. For example, the widths of the windows **123***a*, **123***b*, and **123***c* may be greater than those of the light emitting devices **10***a*, **10***b*, and **10***c*, and thus, a gap may be formed between the light emitting device **10***a* and the light blocking layer **123**.
(87) The adhesive layer **125** is attached onto the transparent substrate **121**. The adhesive layer **125** may cover the light blocking layer **123**, as shown in FIGS. **4**B-**4**C. The adhesive layer **125** may be attached onto an entire surface of the transparent substrate **121**, but the inventive concepts are not

limited thereto, and, in some exemplary embodiments, the adhesive layer **125** may be attached to a portion of the transparent substrate **121** so as to expose a region near an edge of the transparent substrate **121**. The adhesive layer **125** is used to attach the light emitting devices **10***a*, **10***b*, and **10***c* to the transparent substrate **121**. The adhesive layer **125** may fill the window **123***a*, **123***b*, and **123***c* formed in the light blocking layer **123**.
(88) The adhesive layer **125** may be formed as a light transmissive layer, and transmits light emitted from the light emitting devices **10***a*, **10***b*, and **10***c*. The adhesive layer **125** may be formed using an organic adhesive. For example, the adhesive layer **125** may be formed using a transparent epoxy. In addition, the adhesive layer **125** may include a diffuser such as SiO.sub.2, TiO.sub.2, ZnO, or the like to diffuse light. A light diffuser prevents the light emitting devices **10***a*, **10***b* and **10***c* from being observed from the light exiting surface.
(89) Meanwhile, the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* are disposed on the transparent substrate **121**. The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be attached to the transparent substrate **121** by the adhesive layer **125**. The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be disposed corresponding to the windows **123***a*, **123***b*, and **123***c* of the light blocking layer **123**.
(90) The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be disposed on a flat surface of the adhesive layer **125** as shown in FIGS. **4**B and **4**C. The adhesive layer **125** may be disposed under lower surfaces of the light emitting devices **10***a*, **10***b*, and **10***c*, as shown in FIGS. **4**B-**4**C. In another exemplary embodiment, the adhesive layer **125** may partially cover side surfaces of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*.
(91) The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be, for example, a red light emitting device, a green light emitting device, and a blue light emitting device. The detailed configuration of each of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* is described above with reference to FIGS. **3**A and **3**B, and a detailed description thereof will be omitted.
(92) The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be arranged in a line, as illustrated in FIG. **4**A. In particular, in a case that the transparent substrate **121** is a sapphire substrate, the sapphire substrate may include clean-cut surfaces (e.g., m-plane) and non-clean-cut surfaces (e.g., a-plane) due to a location of a crystal plane along a cutting direction. For example, when the sapphire substrate is cut into a quadrangular shape, two cutting planes on both sides thereof (e.g., m-plane) may be cut cleanly along the crystal plane, and two remaining cutting planes (e.g., a-plane) disposed in a direction perpendicular to the cutting planes may not be cut cleanly. In this case, the clean-cut surfaces of the sapphire substrate **121** may be flush with an arrangement direction of the light emitting devices **10***a*, **10***b*, and **10***c*. For example, in FIG. **4**A, the clean-cut surfaces (e.g., m-plane) may be disposed up and down, and the two remaining cut surfaces (e.g., a-plane) may be disposed left and right.
(93) In addition, each of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may be arranged in parallel to one another in a major axis direction. Minor axis directions of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may coincide with an arrangement direction of the light emitting devices.
(94) The first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* may have been those described above with reference to FIGS. **3**A and **3**B, but the inventive concepts are not limited thereto, and various light emitting devices of a lateral type or a flip-chip structure may be used.
(95) As shown in FIGS. **4**B-**4**C, the step adjustment layer **127** covers the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* and the adhesive layer **125**. The step adjustment layer **127** has openings **127***a* exposing the first and second electrode pads **61** and **63** of the light emitting devices **10***a*, **10***b*, and **10***c*. The step adjustment layer **127** assists to securely form the connection layers by uniformly adjusting elevations of surfaces on which the connection layers **129***a*, **129***b*, **129***c*, and **129***d* are formed. The step adjustment layer **127** may be formed of, for example,

photosensitive polyimide.
(96) The step adjustment layer **127** may be disposed in a region surrounded by an edge of the adhesive layer **125**, but the inventive concepts are not limited thereto. For example, the step adjustment layer **127** may be formed to partially expose the edge of the adhesive layer **125**.
(97) A side surface of the step adjustment layer **127** may be inclined at an angle less than 90 degrees with respect to an upper surface of the adhesive layer **125**, as shown in FIG. **4**B. For example, the side surface of the step adjustment layer **127** may have an inclination angle of about 60 degrees with respect to the upper surface of the adhesive layer **125**.
(98) The first, second, third, and fourth connection layers **129***a*, **129***b*, **129***c*, and **129***d* are formed on the step adjustment layer **127**, as shown in FIG. **4**A-**4**C. The connection layers **129***a*, **129***b*, **129***c*, and **129***d* may be connected to the first and second electrode pads **61** and **63** of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c* through the openings **127***a* of the step adjustment layer **127**, as shown in FIG. **4**C.
(99) In an exemplary embodiment, as illustrated in FIGS. **4**A and **4**B, the first connection layer **129***a* may be electrically connected to a second conductivity type semiconductor layer of the first light emitting device **10***a*, the second connection layer **129***b* may be electrically connected to a second conductivity of the second light emitting device **10***b*, the third connection layer **129***c* may be electrically connected to a second conductivity type semiconductor layer of the third light emitting device **10***c*, and the fourth connection layer **129***d* may be commonly electrically connected to first conductivity type semiconductor layers of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*. The first, second, third, and fourth connection layers **129***a*, **129***b*, **129***c*, and **129***d* may be formed together on the step adjustment layer **127**, and may include, for example, Au.
(100) In another exemplary embodiment, the first connection layer **129***a* may be electrically connected to the first conductivity type semiconductor layer of the first light emitting device **10***a*, the second connection layer **129***b* may be electrically connected to the first conductivity type semiconductor layer of the second light emitting device **10***b*, the third connection layer **129***c* may be electrically connected to the first conductivity type semiconductor layer of the third light emitting device **10***c*, and the fourth connection layer **129***d* may be commonly electrically connected to the second conductivity type semiconductor layers of the first, second, and third light emitting devices **10***a*, **10***b*, and **10***c*. The first, second, third, and fourth connection layers **129***a*, **129***b*, **129***c*, and **129***d* may be formed together on the step adjustment layer **127**.
(101) The insulation material layer **131** may be formed to have a thickness smaller than that of the step adjustment layer **127**. A sum of the thicknesses of the insulation material layer **131** and the step adjustment layer **127** may be about 1 μm or more and about 50 μm or less, but the inventive concepts are not limited thereto. Meanwhile, a side surface of the insulation material layer **131** may have an inclination angle less than 90 degrees with respect to the upper surface of the adhesive layer **125**, for example, an inclination angle of about 60 degrees.
(102) The insulation material layer **131** covers side surfaces of the step adjustment layer **127** and the connection layers **129***a*, **129***b*, **129***c*, and **129***d*. In addition, the insulation material layer **131** may cover a portion of the adhesive layer **125** as shown in FIG. **4**C. The insulation material layer **131** may have openings **131***a*, **131***b*, **131***c*, and **131***d* exposing the connection layers **129***a*, **129***b*, **129***c*, and **129***d*, and thus, pad regions of the unit pixel **100** may be defined.
(103) In an exemplary embodiment, the insulation material layer **131** may be a translucent material, and may be formed of an organic or inorganic material. The insulation material layer **131** may be formed of, for example, polyimide. When the insulation material layer **131** along with the step adjustment layer **127** is formed of polyimide, all of lower, side, and upper surfaces of the connection layers **129***a*, **129***b*, **129***c*, and **129***d* may be surrounded by the polyimide, except for the pad regions.
(104) Meanwhile, the unit pixel **100** may be mounted on a circuit board using a bonding material such as solder, and the bonding material may bond the connection layers **129***a*, **129***b*, **129***c*, and

129*d* exposed to the openings **131***a*, **131***b*, **131***c*, and **131***d* of the insulation material layer **131** to pads on the circuit board.

(105) According to the illustrated exemplary embodiment, the unit pixel **100** does not include separate bumps, and the connection layers **129***a*, **129***b*, **129***c*, and **129***d* are used as bonding pads. However, the inventive concepts are not limited thereto, and bonding pads covering the openings **131***a*, **131***b*, **131***c*, and **131***d* of the insulation material layer **131** may be formed. In an exemplary embodiment, the bonding pads may be formed to partially cover the light emitting devices **10***a*, **10***b*, and **10***c* outside of upper regions of the first, second, third, and fourth connection layers **129***a*, **129***b*, **129***c*, and **129***d*.

(106) In the illustrated exemplary embodiment, the light emitting devices **10***a*, **10***b*, and **10***c* are described as being attached to the transparent substrate **121** by the adhesive layer **125**. Alternatively, the light emitting devices **10***a*, **10***b*, and **10***c* may be coupled to the transparent substrate **121** using another coupler instead of the adhesive layer **125**. For example, the light emitting devices **10***a*, **10***b*, and **10***c* may be coupled to the transparent substrate **121** using spacers, and thus, gas or liquid may be filled in a region between the light emitting devices **10***a*, **10***b*, and **10***c* and the transparent substrate **121**. Additionally, an optical layer that transmits light emitted from the light emitting devices **10***a*, **10***b*, and **10***c* may be formed by the gas or liquid. The adhesive layer **125** described above is also an example of the optical layer. Herein, the optical layer is formed of a material such as gas, liquid, or solid, different from those of the light emitting devices **10***a*, **10***b*, and **10***c*, and thus, is distinguished from the materials of the semiconductor layers in the light emitting devices **10***a*, **10***b*, and **10***c*.

(107) FIG. **5**A is a partially enlarged plan view illustrating a portion of FIG. **2** so as to describe the pixel module according to an exemplary embodiment, FIG. **5**B is a schematic partially enlarged plan view illustrating a circuit board according to an exemplary embodiment, and FIG. **5**C is a schematic cross-sectional view taken along line D-D′ of FIG. **5**A. Herein, FIG. **5**B is a plan view illustrating the circuit board **1001** in which the molding member **200** and the unit pixel **100** are omitted from the plan view of FIG. **5**A, but a location of the unit pixel **100** is indicated by a dotted line.

(108) Referring to FIGS. **5**A, **5**B, and **5**C, the pixel module **1000** includes the circuit board **1001** and the unit pixels **100** arranged on the circuit board **1001**. Furthermore, the pixel module **1000** may further include the molding member **200** covering the unit pixels **100**.

(109) The circuit board **1001** may include a base **1010**, a first photosensitive solder resist (PSR) **1030** disposed on the base **1010**, and a second PSR **1050** covering the first PSR **1030**. The circuit board **1001** may include a circuit for electrically connecting the panel substrate **2100** and the light emitting devices **10***a*, **10***b*, and **10***c*. The circuit board **1001** may also include a passive circuit for driving the light emitting devices **10***a*, **10***b*, and **10***c* in a passive matrix driving manner or an active circuit for driving the light emitting devices **10***a*, **10***b*, and **10***c* in an active matrix driving manner. The circuits in the circuit board **1001** may be formed in a multi-layer structure in the base **1010**, and some circuits are shown in FIG. **5**C. Meanwhile, interconnections **1003** are disposed on an upper surface of the base **1010**.

(110) The first PSR **1030** covers the interconnections **1003** and has a pad open region **1030***a* defining first portions **1003***a* of the interconnections. The first PSR **1030** is applied so as to cover the base **1010** and then patterned through photolithography and development processes. For example, the first PSR **1030** may be formed using an ultraviolet curable resin, a remaining region except for the pad open region **1030***a* is irradiated with ultraviolet light, and an ultraviolet curable resin of the pad open region **1030***a* may be removed through the development process. The first PSR **1030** may be formed to have a relatively thin thickness so as to clearly define the pad open region **1030***a*. The first PSR **1030** may be formed on the interconnections **1003** to have a thickness of less than 20 μm, and may be formed on the interconnections **1003** to have a thickness of, for example, about 10 μm to about 15 μm.

(111) The first portions **1003***a* of the interconnections **1003** are exposed through the pad open region **1030***a*. Meanwhile, remaining portions **1003***b* and **1003***c* of the interconnections **1003** are covered with the first PSR **1030**. Since the first PSR **1030** has a relatively thin thickness, the remaining portions **1003***b* and **1003***c* of the interconnections may be observed through the first PSR **1030**.
(112) Meanwhile, the second PSR **1050** covers the first PSR **1030**. The second PSR **1050** has an opening **1050***a* exposing the pad open region **1030***a*. The opening **1050***a* is larger than the pad open region **1030***a*, and the pad open region **1030***a* is exposed through the opening **1050***a*. An area of the first PSR **1030** exposed in the opening **1050***a* of the second PSR **1050** may be 50% or less of the pad open region **1030***a*. A ratio of the exposed area of the first PSR **1030** to the pad open region **1030***a* decreases as a size of the unit pixel **100** decreases. The second PSR **1050** may be applied so as to cover the first PSR **1030** and then patterned through the photolithography and development processes. The opening **1050***a* of the second PSR **1050** does not need to be as clear as the pad open region **1030***a*, and thus, a thickness of the second PSR **1050** may be identical to or larger than that of the first PSR **1030** on the interconnections **1003**. The second PSR **1050** may have a thickness of 20 μm or more, for example, about 20 μm to about 25 μm.
(113) The first PSR **1030** and the second PSR **1050** may be formed of the same material, but the inventive concepts are not limited thereto. The first PSR **1030** and the second PSR **1050** may include an ultraviolet curable resin and a light absorbing material dispersed in the ultraviolet curable resin. The light absorbing material is a material that absorbs visible light and may be carbon black, for example. A light transmittance of the first PSR **1030** and a light transmittance of the second PSR **1050** may be adjusted using the light absorbing material. However, since excessive addition of the light absorbing material hinders resin curing by ultraviolet rays, an amount of the light absorbing material added into the ultraviolet curable resin is limited. For this reason, it is difficult to form the first PSR **1030** or the second PSR **1050** with a material having a low light transmittance.
(114) When a single-layered PSR is used, it is difficult to lower the light transmittance of the PSR while clearly defining the pad open region. That is, if the thickness of the PSR is reduced, the pad open region is clearly defined, but as the interconnection **1003** is observed through the PSR, it is difficult to implement black color, and if the thickness of the PSR is increased, it is difficult to clearly define the pad open region. In contrast, in the present disclosure, the second PSR **1050** is additionally formed while the pad open region **1030***a* is clearly defined using the first PSR **1030**, an observation of the interconnection **1003** may be reduced, and thus black color may be displayed better.
(115) As it can be seen from FIG. **5**B, the interconnection **1003** may include a first portion **1003***a* exposed to the pad open region **1030***a*, a second portion **1003***b* covered with the first PSR **1030**, and a third portion **1003***c* covered with the first PSR **1030** and the second PSR **1050**. Herein, as the unit pixel **100** is disposed in the pad open region **1030***a*, the first portion **1003***a* is covered by the unit pixel **100**. In addition, as the third portion **1003***c* is covered with the first PSR **1030** and the second PSR **1050**, light is blocked by the entire thicknesses of the first PSR **1030** and the second PSR **1050**, and thus it is difficult to observe the third portion **1003***c* externally. Meanwhile, an area of the unit pixel **100** may be larger than that of the pad open region **1030***a* and smaller than that of the opening **1050***a* of the second PSR **1050**. Accordingly, a portion of the first PSR **1030** exposed in the opening **1050***a* overlaps the unit pixel **100**. In addition, a portion of the second portion **1003***b* of the interconnection **1003** overlaps the unit pixel **100**. Meanwhile, at least portions of side surfaces of the unit pixels **100** may be located inside the opening **1050***a* of the second PSR **1050**. Accordingly, when the unit pixels **100** move in a lateral direction while the unit pixels **100** are mounted, the side surface of the unit pixel **100** may contact a side wall of the opening **1050***a* of the second PSR **1050** and thus, it is possible to prevent the unit pixel **100** from being deviated from the opening **1050***a*.

(116) Meanwhile, as a portion among the second portion **1003***b* of the interconnection **1003** disposed outside the unit pixel **100** is covered only by the first PSR **1030**, the second portion **1003***b* may be observed externally. However, the second portion **1003***b* is located in a region between the pad open region **1030***a* and the opening **1050***a* and furthermore, a part of the second portion **1003***b* is covered by the unit pixel **100**, and therefore, the interconnection portion that can be observed from the outside is limited to an extremely small region.
(117) The detailed configuration of the unit pixels **100** is described above with reference to FIGS. **4**A, **4**B, and **4**C. The unit pixels **100** may be arranged on the circuit board **1001**. The unit pixels **100** may be arranged in various matrices, such as 2×2, 2×3, 3×3, 4×4, 5×5, and the like. Examples in which the unit pixels **100** are arranged in matrices of 2×2 and 3×2 will be described later with reference to FIGS. **8**A through **9**C.
(118) As shown in FIG. **5**C, the unit pixels **100** may be bonded to the circuit board **1001** through a bonding material **1005**. For example, the bonding material **1005** bonds, to the first portions on the circuit board **1001** (i.e., pads **1003***a*), connection layers **229***a*, **229***b*, **229***c*, and **229***d* exposed through the openings **131***a*, **131***b*, **131***c*, and **131***d* of the insulation material layer **131** described with reference to FIGS. **4**A, **4**B, and **4**C. For example, the bonding material **1005** may be solder, and after a solder paste is disposed on the pads **1003***a* on the circuit board **1001** using a technology such as screen printing, the unit pixel **100** and the circuit board **1001** may be bonded through a reflow process. The pads **1003***a* on the circuit board **1001** are disposed under the first and second PSRs **1030** and **1050**, and an upper surface of the unit pixel **100** may protrude above an upper surface of the second PSR **1050**.
(119) According to the illustrated exemplary embodiment, the bonding material **1005** having a single structure may be disposed between the connection layers **129***a*, **129***b*, **129***c*, and **129***d* and the pads **1003***a*, and the bonding material **1005** may directly connect the connection layers **129***a*, **129***b*, **129***c*, and **129***d* and the pads **1003***a*. Alternatively, metallic bumps may be formed on the connection layers **129***a*, **129***b*, **129***c* and **129***d*, and the metallic bumps may be bonded to the pads **1003***a* through the bonding material **1005**.
(120) The molding member **200** covers the plurality of unit pixels **100**. A total thickness of the molding member **200** may be in a range of about 150 um to 350 um. The molding member **200** may be a single layer, without being limited thereto, but it may be a multilayer. The molding member **200** may be a transparent or opaque molding member. Furthermore, the molding member **200** may further include a light diffusion layer. The light diffusion layer may include a transparent matrix such as an epoxy molding compound and light diffusion particles dispersed in the transparent matrix. The light diffusion particles may be, for example, silica or TiO.sub.2, without being limited thereto.
(121) The opaque molding member includes, for example, a transparent matrix and a material that absorbs light in the transparent matrix. The matrix may be, for example, a dry-film type solder resist (DFSR), photoimageable solder resist (PSR), an epoxy molding compound (EMC), or the like, without being limited thereto. The light absorbing material may include a light absorbing dye such as carbon black. The light absorbing dye may be directly dispersed in the matrix, or may be coated on surfaces of organic or inorganic particles to be dispersed in the matrix. Various types of organic or inorganic particles may be used so as to coat the light absorbing material. For example, particles coated with TiO.sub.2 or silica particles with carbon black may be used. A light transmittance may be adjusted by adjusting a concentration of the light absorbing material contained in the molding member **200**.
(122) The molding member **200** may be formed as a single layer in which the light absorbing material is uniformly dispersed, but the inventive concepts are not limited thereto. The molding member **200** may be formed of a plurality of layers having different concentrations of the light absorbing material. For example, the molding member **200** may include two layers having different concentrations of the light absorbing material.

(123) In an exemplary embodiment, when the molding member **200** is formed of the plurality of layers, the layers may be clearly distinguished from one another. For example, after the layers having different concentrations of the light absorbing material are individually manufactured as films, the molding member **200** may be manufactured by sandwiching the films. Alternatively, the molding member **200** may be formed by continuously printing the layers having different concentrations of the light absorbing material. In another exemplary embodiment, the molding member **200** may be formed such that the concentration of the light absorbing material gradually decreases in a thickness direction thereof.
(124) Light incident perpendicularly from the unit pixels **100** has a short path passing through the molding member **200** and thus, easily passes through the molding member **200**, but light incident with an inclination angle has a long path through the molding member **200**. As such, when the molding member **200** is formed of a black molding member, light interference between the unit pixels **100** may be prevented, and a contrast of the displaying apparatus may be improved and a color deviation may be reduced.
(125) The molding member **200** may be formed using, for example, the technique such as lamination, spin coating, slit coating, printing, or the like. As an example, the molding member **200** may be formed on the unit pixels **100** by vacuum lamination using a molding member in a form of a film or a sheet.
(126) A displaying apparatus **10000** may be provided by mounting the plurality of pixel modules **1000** on the panel substrate **2100** of FIG. **1**. The circuit board **1001** has bottom pads connected to the pads **1003***a*. The bottom pads may be disposed to correspond one-to-one to the pads **1003***a*, but the number of the bottom pads may be reduced through a common connection.
(127) In the illustrated exemplary embodiment, the unit pixels **100** are formed into the pixel module **1000**, the pixel modules **1000** are mounted on the panel substrate **2100**, and the displaying apparatus **10000** may be provided, and thus, a process yield of the displaying apparatus may be improved. However, the inventive concepts are not limited thereto, and the unit pixels **100** may be directly mounted on the panel substrate **2100**.
(128) FIG. **6**A is a schematic cross-sectional view illustrating a unit pixel **100***a* according to another exemplary embodiment, and FIG. **6**B is a schematic plan view illustrating the unit pixel **100***a* according to FIG. **6**A.
(129) Referring to FIGS. **6**A and **6**B, the unit pixel **100***a* has a structure in which first, second, and third light emitting stacks **320**, **330**, and **340** are stacked, unlike the unit pixel **100** described with reference to FIGS. **4**A, **4**B, and **4**C.
(130) The unit pixel **100***a* includes a light emitting stacked structure, a first connection electrode **350***a*, a second connection electrode **350***b*, a third connection electrode **350***c*, and a fourth connection electrode **350***d* formed on the light emitting stacked structure, and a passivation layer **390** surrounding the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d*. The unit pixel **100***a* may also include a substrate **311**. Meanwhile, the light emitting stacked structure may include the first light emitting stack **320**, the second light emitting stack **330**, and the third light emitting stack **340**. Although the light emitting stacked structure has been illustrated as being configured to include the three light emitting stacks **320**, **330**, and **340**, the inventive concepts are not limited to a specific number of light emitting stacks. For example, in some exemplary embodiments, the light emitting stacked structure may include two or more light emitting stacks. Herein, it will be described that the unit pixel **100***a* includes the three light emitting stacks **320**, **330**, and **340** according to an exemplary embodiment.
(131) The substrate **311** may be a light transmissive insulating substrate. However, in some exemplary embodiments, the substrate **311** may be formed to be translucent or partially transparent so as to transmit only light of a specific wavelength or transmit only a portion of light of a specific wavelength. The substrate **311** may be a growth substrate on which the first light emitting stack **320** may be epitaxially grown, for example, a sapphire substrate. However, the substrate **311** is not

limited to the sapphire substrate, and may include various other transparent insulating materials. For example, the substrate **311** may include a glass, a quartz, a silicon, an organic polymer, or an organic-inorganic composite material, and may be, for example, such as silicon carbide (SiC), gallium nitride (GaN), indium gallium nitride (InGaN), aluminum gallium nitride (AlGaN), aluminum nitride (AlN), gallium oxide (Ga.sub.2O.sub.3), or a silicon substrate. In addition, the substrate **311** may include irregularities on an upper surface thereof, and may be, for example, a patterned sapphire substrate. By including the irregularities on the upper surface, extraction efficiency of light generated in the first light emitting stack **320** in contact with the substrate **311** may be increased. The irregularities of the substrate **311** may be employed so as to selectively increase a luminous intensity of the first LED stack **320** compared to the second LED stack **330** and the third LED stack **340**.

(132) The first, second, and third light emitting stacks **320**, **330**, and **340** are configured to emit light towards the substrate **311**. Accordingly, light emitted from the third light emitting stack **340** may pass through the first and second light emitting stacks **320** and **330**. According to an exemplary embodiment, the first, second, and third light emitting stacks **320**, **330**, and **340** may emit light of different peak wavelengths from one another. In general, the light emitting stack disposed farther from the substrate **311** emits light of a longer wavelength than that of the light emitting stack disposed near the substrate **311**, thereby reducing a light loss. However, in the present disclosure, so as to adjust a color mixing ratio of the first, second, and third light emitting stacks **320**, **330**, and **340**, the second LED stack **330** may emit light of a shorter wavelength than that of the first LED stack **320**. Accordingly, it is possible to reduce the luminous intensity of the second LED stack **330** and increase the luminous intensity of the first LED stack **320**, and thus, it is possible to dramatically change a luminous intensity ratio of light emitted from the first, second, and third light emitting stacks. For example, the first light emitting stack **320** may be configured to emit green light, the second light emitting stack **330** to emit blue light, and the third light emitting stack **340** to emit red light. Accordingly, it is possible to relatively decrease the luminous intensity of blue light and relatively increase the luminous intensity of green light. As a result, the luminous intensity ratio of red, green, and blue may be adjusted to be close to 3:6:1. Furthermore, an emission area of the first, second, and third LED stacks **320**, **330**, and **340** may be less than or equal to about 10000 μm.sup.2, specifically, may be less than or equal to 4000 μm.sup.2. Additionally, or alternatively, an emission area of the first, second, and third LED stacks **320**, **330**, and **340** may be less than or equal to 2500 μm.sup.2. In addition, the emission area may be increased with a decreasing distance to the substrate **311**, and the luminous intensity of green light may be further increased by disposing the first LED stack **320** emitting green light closest to the substrate **311**.

(133) The first to the third light emitting stacks **320**, **330**, and **340** include, as those described with reference to FIGS. **3**A and **3**B, a first conductivity type semiconductor layer **21**, an active layer **23**, and a second conductivity type semiconductor layer **25**, respectively. According to an exemplary embodiment, the first light emitting stack **320** may include a semiconductor material emitting green light, such as GaN, InGaN, GaP, AlGaInP, AlGaP, or the like. The second light emitting stack **330** may include a semiconductor material emitting blue light, such as GaN, InGaN, ZnSe, or the like, without being limited thereto. According to an exemplary embodiment, the third light emitting stack **340** may include, for example, a semiconductor material emitting red light such as AlGaAs, GaAsP, AlGaInP, GaP, or the like, without being limited thereto.

(134) According to an exemplary embodiment, each of the first conductivity type semiconductor layers **21** and the second conductivity type semiconductor layers **25** of the first, second, and third light emitting stacks **320**, **330**, and **340** may have a single-layered structure or a multi-layered structure and, in some exemplary embodiments, may include a superlattice layer. Furthermore, the active layers **23** of the first, second, and third light emitting stacks **320**, **330**, and **340** may have a single quantum well structure or a multiple quantum well structure.

(135) A first adhesive layer **325** is disposed between the first light emitting stack **320** and the second light emitting stack **330**, and a second adhesive layer **335** is disposed between the second light emitting stack **330** and the third light emitting stack **340**. The first and second adhesive layers **325** and **335** may include a non-conductive material that transmits light. For example, the first and second adhesive layers **325** and **335** may include an optically clear adhesive (OCA), which may include epoxy, polyimide, SUB, spin-on-glass (SOG), benzocyclobutene (BCB), without being limited thereto.
(136) According to an exemplary embodiment, each of the first, second, and third light emitting stacks **320**, **330**, and **340** may be driven independently. More specifically, a common voltage may be applied to one of the first and second conductivity type semiconductor layers of each of the light emitting stacks, and an individual light emitting signal may be applied to another one of the first and second conductivity type semiconductor layers of each of the light emitting stacks. According to an exemplary embodiment of the present disclosure, the first conductivity type semiconductor layer **21** of each of the light emitting stacks may be n-type, and the second conductivity type semiconductor layer **25** may be p-type. In the first light emitting stack **320**, the second light emitting stack **330**, and the third light emitting stack **340**, the n-type semiconductor layer and the p-type semiconductor layer may be arranged in the same sequence, but the inventive concepts are not limited thereto. For example, the first light emitting stack **320** may have a reversely stacked sequence compared to those of the second light emitting stack **330** and the third light emitting stack **340**. The first, second, and third light emitting stacks **320**, **330**, and **340** may have a common p-type light emitting stacked structure in which the p-type semiconductor layers are commonly electrically connected, or may have a common n-type light emitting stacked structure in which the n-type semiconductor layers are commonly electrically connected.
(137) According to the illustrated exemplary embodiment, each of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may have a substantially elongated shape protruding from the substrate **311**. The connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may include a metal such as Cu, Ni, Ti, Sb, Zn, Mo, Co, Sn, Ag, or an alloy thereof, without being limited thereto. For example, each of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may include two or more metals or a plurality of different metallic layers so as to reduce stress from the elongated shape of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d*. In another exemplary embodiment, when the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* include Cu, an additional metal may be deposited or plated to suppress oxidation of Cu. In some exemplary embodiments, when the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* include Cu/Ni/Sn, Cu may prevent Sn from infiltrating into the light emitting stacked structure. In some exemplary embodiments, the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may include a seed layer for forming a metallic layer during a plating process, which will be described later.
(138) As shown in FIG. **6**B, each of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may have a substantially flat upper surface, and thus, an electrical connection between an external line or an electrode and the light emitting stacked structure may be facilitated, which will be described later. According to an exemplary embodiment of the present disclosure, when the unit pixel **100***a* is a micro-LED, which has a surface area less than about 10,000 μm.sup.2 as known in the art, the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may overlap a portion of at least one of the first, second, and third light emitting stacks **320**, **330**, and **340** as shown in FIGS. **6**A-**6**B. The unit pixel **100***a* has a surface area less than about 4,000 μm.sup.2 or 2,500 μm.sup.2 in other exemplary embodiments. In some forms, the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* are illustrated as having a quadrangular pillar shape, but the present disclosure is not limited thereto, and other shapes such as a cylindrical shape are possible. Furthermore, areas of lower surfaces of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* may be larger than those of the upper surfaces thereof. For example, when the first to third light emitting stacks **320**, **330**, and **340** are patterned so as to form electrodes, the connection electrodes **350***a*, **350***b*, **350***c* and **350***d* may cover side

surfaces of the first to third light emitting stacks **320**, **330**, and **340**.
(139) In general, during manufacturing, an array of a plurality of unit pixels **100***a* is formed on the substrate **311**. The substrate **311** may be cut along scribing lines to be singularized (isolated) into each unit pixel **100***a*, and the unit pixel **100***a* may be transferred to another substrate or tape using various transferring techniques. In this case, when the unit pixel **100***a* includes the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* such as metallic bumps or pillars protruding outward, various drawbacks may occur during subsequent processes, for example, in a transferring step, due to the structure in which the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* are exposed to the outside. Moreover, when the unit pixel **100***a* includes a micro-LED, which has a surface area less than about 10,000 μm.sup.2, such as less than about 4,000 μ.sup.2, or less than about 2,500 μm.sup.2, depending upon applications, handling of the unit pixel **100***a* may become more difficult due to its small form factor.
(140) For example, when the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* have a substantially elongated shape such as a rod, transferring the unit pixel **100***a* using a conventional vacuum method is difficult due to a protruding structure of the connection electrode due to an insufficient suction area. In addition, the exposed connection electrode may be directly affected by various stresses during subsequent processes, such as when the connection electrode is in contact with a manufacturing device, which may damage the structure of the unit pixel **100***a*. As another example, by attaching an adhesive tape on an upper surface (e.g., a surface opposite to the substrate **311**) of the unit pixel **100***a*, a contact area between the unit pixel **100***a* and the adhesive tape may be limited to the upper surfaces of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* when the unit pixel **100***a* is transferred. In this case, unlike when the adhesive tape is attached to a lower surface of the substrate, an adhesive force of the unit pixel **100***a* to the adhesive tape may be weakened, and the unit pixel **100***a* may be undesirably separated from the adhesive tape during transferring. As yet another example, when the unit pixel **100***a* is transferred using a conventional pick-and-place method, an ejection pin may directly contact a portion of the unit pixel **100***a* and damage a top structure of the light emitting structure. In particular, the ejection pin may strike a center of the unit pixel **100***a* and cause physical damage to the top light emitting stack of the unit pixel **100***a*.
(141) According to an exemplary embodiment of the present disclosure, the passivation layer **390** may be formed on the light emitting stacked structure. More specifically, as shown in FIG. **6**A, the passivation layer **390** is formed between the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d* to cover side surfaces of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d*. Furthermore, although the passivation layer **390** has described as being disposed on the light emitting stacked structure in FIGS. **6**A-**6**B, the passivation layer **390** may at least partially cover the side surfaces of the first to third light emitting stacks **320**, **330**, and **340**, and the side surfaces of the first to third light emitting stacks **320**, **330**, and **340** may not be exposed to the outside of the unit pixel **100***a* by being covered with the passivation layer **390** and another insulation layer.
(142) The passivation layer **390** may be formed substantially flush with the upper surfaces of the connection electrodes **350***a*, **350***b*, **350***c*, and **350***d*. The passivation layer **390** may include an epoxy molding compound (EMC), which may be formed in various colors such as black, white, or transparent. However, the inventive concepts are not limited thereto. For example, in some exemplary embodiments, the passivation layer **390** may include polyimide (PID), and in this case, the PID may be provided as a dry film rather than a liquid type so as to increase a level of flatness when applied to the light emitting stacked structure. In some exemplary embodiments, the passivation layer **390** may include a photosensitive material. In this manner, the passivation layer **390** may protect the light emitting stacked structure from an external impact that may be applied during subsequent processes, as well as providing a sufficient contact area to the unit pixel **100***a* so as to facilitate its handling during subsequent transferring steps. In addition, the passivation layer **390** may prevent light leakage toward the side surface of the unit pixel **100***a* so as to prevent or at

least suppress interference of light emitted from adjacent unit pixels **100***a*.
(143) The unit pixel **100***a* according to the illustrated exemplary embodiment may be formed to have a size smaller than that of the unit pixel **100** including the light emitting devices **10***a*, **10***b*, and **10***c* arranged on the same plane as the unit pixel **100***a* has a structure in which the first, second, and third light emitting stacks **320**, **330**, and **340** are stacked. For example, the unit pixel **100** may have a size of 400 μm×400 μm, and the unit pixel **100***a* according to the illustrated exemplary embodiment may have a size of 225 μm×225 μm.
(144) FIG. **7**A is an image illustrating a pixel module according to the related art, and FIG. **7**B is an image illustrating a pixel module according to an exemplary embodiment. The pixel module according to the related art uses only a first PSR **1030** to cover an interconnection **1003**, and the pixel module according to the present exemplary embodiment uses a second PSR **1050** together with the first PSR **1030**.
(145) As it can be seen from FIG. **7**A, when only the first PSR **1030** is used, the interconnection **1003** covered with the first PSR **1030** is clearly observed. In contrast, as it can be seen from FIG. **7**B, the interconnection **1003** covered with the first PSR **1030** and the second PSR **1050** is faintly observed, and a portion of the interconnection **1003** covered only with the first PSR **1030** is relatively very small region.
(146) In the pixel module according to an exemplary embodiment, the unit pixel **100** has an area of 400 μm×400 μm, an area of a pad open region of the first PSR **1030** is about 122,500 μm.sup.2, and an area of an opening of the second PSR **1050** is about 176,400 μm.sup.2. An area of the first PSR **1030** exposed through the opening of the second PSR **1050** is about 44% of the area of the pad open region of the first PSR **1030**.
(147) Meanwhile, in an example in which the pixel is manufactured using the unit pixel **100***a* described with reference to FIGS. **6**A and **6**B, the unit pixel **100***a* has an area of 225 μm×225 μm, an area of a pad open region of the first PSR **1030** was about 46,225 μm.sup.2, and an area of the opening of the second PSR **1050** was about 60,025 μm.sup.2. In this case, the area of the first PSR **1030** exposed through the opening of the second PSR **1050** is about 30% of the area of the pad open region of the first PSR **1030**.
(148) FIG. **8**A is a schematic plan view illustrating a pixel module **1000***a* according to an exemplary embodiment, FIG. **8**B is a schematic rear view illustrating the pixel module **1000***a* according to an exemplary embodiment, and FIG. **8**C is a schematic circuit diagram illustrating the pixel module **1000***a* according to an exemplary embodiment.
(149) Referring to FIGS. **8**A, **8**B, and **8**C, the pixel module **1000***a* is substantially similar to the pixel module **1000** described above, except that the unit pixels **100** are arranged in a 2×2 matrix. That is, four unit pixels **100** are mounted on the circuit board **1001**.
(150) Four first portions **1003***a* corresponding to each of the unit pixels **100** are exposed by the pad open region **1030***a* of the first PSR **1030**. Meanwhile, the opening **1050***a* of the second PSR **1050** has a size larger than that of the pad open region **1030***a* and exposes the pad open region **1030***a*. A portion of the first PSR **1030** around the pad open region **1030***a* is exposed in the opening **1050***a*.
(151) The unit pixels **100** are mounted in the opening of the second PSR **1050**. The unit pixel **100** has an area larger than that of the pad open region **1030***a*, and covers the first portions **1003***a* to block the first portions **1003***a* from being observed from the outside.
(152) Meanwhile, terminals C**1**, C**2**, R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** for electrical connection are disposed on a rear surface of the circuit board **1001**. These terminals C**1**, C**2**, R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** are electrically connected to the unit pixels **100** through the interconnections **1003** in the circuit board **1001**. Multi-layered interconnections **1003** may be disposed in the circuit board **1001**. The interconnections of each layer may be connected to the interconnections of other layers through vias. The terminals C**1**, C**2**, R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** disposed on the rear surface are electrically connected to the first portions **1003***a* disposed on an upper surface through the vias and the interconnections in the circuit board **1001**.

(153) As shown in FIG. **8**C, common terminals C**1** and C**2** are electrically connected to the sub-pixels R, G, and B of two unit pixels **100**, respectively. Meanwhile, individual terminals R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** may be electrically connected to same sub-pixels R, G, and B of two unit pixels **100**, respectively. For example, the individual terminal R**1** may be commonly electrically connected to the sub-pixels R of two unit pixels **100** arranged in a first column, and the individual terminal R**2** may be commonly electrically connected to the sub-pixels R of two unit pixels **100** arranged in a second column. Similarly, the individual terminal G**1** may be commonly electrically connected to the sub-pixels G of two unit pixels **100** arranged in the first column, and the individual terminal G**2** may be commonly electrically connected to the sub-pixels G of two unit pixels **100** arranged in the second column. In addition, the individual terminal B**1** may be commonly electrically connected to the sub-pixels B of two unit pixels **100** arranged in the first column, and the individual terminal B**2** may be commonly electrically connected to the sub-pixels B of two unit pixels **100** arranged in the second column.
(154) Accordingly, the sub-pixels R, G, and B in the four unit pixels **100** may be independently driven using the eight terminals. Although the number of the first portions **1003***a* exposed on the upper surface of the circuit board **1001** is 16, the number of the terminals may be reduced to ½ of the number of the first portions **1003***a*.
(155) FIG. **9**A is a schematic plan view illustrating a pixel module **1000***b* according to another exemplary embodiment, FIG. **9**B is a schematic rear view illustrating the pixel module **1000***b* according to another exemplary embodiment, and FIG. **9**C is a schematic circuit diagram illustrating the pixel module **1000***b* according to another exemplary embodiment.
(156) Referring to FIGS. **9**A, **9**B, and **9**C, the pixel module **1000***b* is substantially similar to the pixel module **1000***a* described above, except that the unit pixels **100** are arranged in a 3×2 matrix. Six unit pixels **100** are mounted on the circuit board **1001**.
(157) Twenty-four first portions **1003***a* on the upper surface of the circuit board **1001** are exposed to the pad open region **1030***a* of the first PSR **1030**, and six unit pixels **100** are mounted in the openings **1050***a* of the second PSR **1050**.
(158) Meanwhile, nine terminals C**1**, C**2**, C**3**, R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** may be disposed on the rear surface of the circuit board **1001**. Multi-layered interconnections **1003** may be disposed in the circuit board **1001**, and the interconnections of each layer may be connected to the interconnections of other layers through vias. The nine terminals C**1**, C**2**, C**3**, R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** disposed on the rear surface are electrically connected to the first portions **1003***a* disposed on the upper surface through the vias and the interconnections in the circuit board **1001**.
(159) Referring to FIG. **9**C, common terminals C**1**, C**2**, and C**3** are commonly electrically connected to the sub-pixels R, G, and B of two unit pixels **100**, respectively. Meanwhile, individual terminals R**1**, R**2**, G**1**, G**2**, B**1**, and B**2** may be commonly electrically connected to same sub-pixels R, G, and B of three unit pixels **100**, respectively. For example, the individual terminal R**1** may be commonly electrically connected to the sub-pixels R of three unit pixels **100** arranged in the first column, and the individual terminal R**2** may be commonly electrically connected to the sub-pixels R of three unit pixels **100** arranged in the second column. Similarly, the individual terminal G**1** may be commonly electrically connected to the sub-pixels G of three unit pixels **100** arranged in the first column, and the individual terminal G**2** may be commonly electrically connected to the sub-pixels G of three unit pixels **100** arranged in the second column. In addition, the individual terminal B**1** may be commonly electrically connected to the sub-pixels B of three unit pixels **100** arranged in the first column, and the individual terminal B**2** may be commonly electrically connected to the sub-pixels B of three unit pixels **100** arranged in the second column.
(160) Accordingly, the sub-pixels R, G, and B in the six unit pixels **100** may be independently driven using the nine terminals. Although the number of the first portions **1003***a* exposed on the upper surface of the circuit board **1001** is 24, the number of the terminals may be reduced to less than ½ of the number of the first portions **1003***a*.

(161) In the illustrated exemplary embodiment, although the unit pixels **100** arranged in the 3×2 matrix have been exemplarily described, it will be easy to understand even when they are arranged in a 2×3 matrix, six unit pixels **100** may be independently driven using the same number of terminals.
(162) Although some embodiments have been described herein, it should be understood that these embodiments are provided for illustration only and are not to be construed in any way as limiting the present disclosure. It should be understood that features or components of an exemplary embodiment can also be applied to other embodiments without departing from the spirit and scope of the present disclosure.

Claims

1. A circuit board, comprising: a base; a plurality of interconnections disposed on an upper surface of the base, each interconnection comprising a plurality of portions; a first photosensitive solder resist (PSR) covering a portion of each of the plurality of interconnections, the first PSR arranged to define a pad open region configured to expose a remainder portion of each of the plurality of interconnection; and a second PSR covering the first PSR and having an opening configured to expose the pad open region, wherein the opening of the second PSR is larger than the pad open region of the first PSR.
2. The circuit board of claim 1, wherein the second PSR is thicker than the first PSR.
3. The circuit board of claim 2, wherein: the first PSR has a thickness of about 10 μm to about 15 μm, and the second PSR has a thickness of about 20 μm to about 25 μm.
4. The circuit board of claim 1, wherein the plurality of interconnections includes four interconnections disposed on the upper surface of the base and the first PSR covers the portion of each of the four interconnections; and the pad open region is further configured to expose the remainder portion of each of the four interconnections.
5. The circuit board of claim 1, wherein: each of the plurality of interconnections includes a first portion located in the pad open region, a second portion located in the opening of the second PSR and covered with the first PSR, and a third portion covered with both the first PSR and second PSR; and the remainder portion of each of the plurality of interconnections corresponds to the first portion and the portion of each of the plurality of interconnections, covered by the first PSR, correspond to the second portion and the third portion.
6. The circuit board of claim 1, wherein the circuit board is a printed circuit board.
7. A displaying apparatus, comprising: a circuit board; a unit pixel disposed on the circuit board; and a molding member covering the unit pixel; the circuit board comprising: a base having a plurality of interconnections on an upper surface thereof; a first photosensitive solder resist (PSR) covering a portion of each of the plurality of interconnection, the first PSR arranged to define a pad open region configured to expose a remainder portion of each of the plurality of interconnections; and a second PSR covering the first PSR and having an opening configured to expose the pad open region, wherein: the opening of the second PSR is larger than the pad open region of the first PSR, and the unit pixel is disposed in the pad open region and in the opening of the second PSR such that the unit pixel is electrically coupled to the remainder portion of each of the plurality of interconnections in the pad open region.
8. The displaying apparatus of claim 7, wherein the unit pixel includes at least three light emitting devices disposed adjacent to one another.
9. The displaying apparatus of claim 7, the unit pixel comprising: a plurality of light emitting stacks stacked one above another; and a plurality of connection electrodes electrically connected to the plurality of light emitting stacks.
10. The displaying apparatus of claim 7, wherein the unit pixel is arranged to cover the pad open region.

11. The displaying apparatus of claim 7, wherein the molding member includes a light absorbing material.
12. The displaying apparatus of claim 7, further comprising: a display panel; and a pixel module disposed on the display panel, wherein the pixel module includes the circuit board and a plurality of unit pixels.
13. A displaying apparatus, comprising: a panel substrate; a plurality of pixel modules arranged on the panel substrate, each pixel module comprising: a circuit board; a plurality of unit pixels disposed on the circuit board; and a molding member covering the plurality of unit pixels; wherein the circuit board further comprises: a base having a plurality of interconnections on an upper surface thereof, wherein an interconnection of the plurality of interconnections comprises a first part, a second part, and a third part; a pad open region formed in a first photosensitive solder resist (PSR) layer, the pad open region configured to expose the first part of the interconnection, and the first PSR layer covering the second part and the third part of the interconnection; and an opening formed in a second PSR layer covering a portion of the first PSR layer, the pad open region exposed in the opening of the second PSR layer; and wherein a unit pixel of the plurality of unit pixels is disposed in the opening such that the unit pixel is electrically connected to the first part of the interconnection in the pad open region.
14. The displaying apparatus of claim 13, wherein the unit pixel covers the pad open region and includes at least a portion mounted in the opening.
15. The displaying apparatus of claim 13, wherein the first part of the interconnection is located in the pad open region, the second part located in the opening of the second PSR layer and covered with the first PSR layer, and the third part covered with both the first PSR layer and the second PSR layer.
16. The displaying apparatus of claim 13, wherein the unit pixel further comprises: a plurality of light emitting stacks stacked one above another on a substrate; and a plurality of connection electrodes electrically connected to the plurality of light emitting stacks.
17. The displaying apparatus of claim 16, wherein: the plurality of light emitting stacks is operable to emit light of different peak wavelengths from one another; and one of the plurality of light emitting stacks, farther from the substrate, emits light of a longer wavelength than another light emitting stack near the substrate.
18. The displaying apparatus of claim 17, wherein an emission area of each of the plurality of light emitting stacks increases as a distance to the substrate decreases.
19. The displaying apparatus of claim 13, wherein: the unit pixel further comprises a plurality of connection electrodes which is electrically connected to the first part of the interconnection via a bonding material; and the bonding material is disposed in the pad open region.